

Run 02 | OME

Global (including India)

IV only

Target Opportunity-motivated entrepreneurship; binary 1 vs. 0
Sample 1,342,884 respondents **Coverage** 89 countries; 2015–2022
Positive cases 100,087 (7.45%) **Case time** 45m 26s
Train / val. / test 913,163 / 161,145 / 268,576 (68% / 12% / 20%, stratified)
Predictors (13): Self-efficacy, opportunity perception, fear of failure, knowing entrepreneurs, intention, media coverage, business exit, business angel, established business owner, good career choice, ease of starting a business, high status, social enterprise.

Legatum predictors (13): Prosperity index score, Safety and security, Personal freedom, Governance, Social capital, Investment environment, Enterprise conditions, Market access and infrastructure, Economic quality, Living conditions, Health, Education, Natural environment. Total independent variables: 26.

Controls (0): None. Country and respondent ID are excluded as predictors.

Protocol: Ten CPU models; model seed 20260925; fixed splits by sample/outcome; training-only preprocessing; F1 threshold selected on validation data. Test scores below are unweighted.

Result in brief

Highest observed ROC-AUC: XGBoost (0.9760). Highest F1: XGBoost (0.6958). The all-negative accuracy baseline is 92.55% (full sample).

Model performance on held-out test records

Model	ROC-AUC	PR-AUC	Acc.%	Prec.%	Recall%	F1	Time
XGBoost	0.9760	0.7093	94.41	58.56	85.73	0.6958	26s
Random Forest	0.9757	0.7117	94.42	58.66	85.25	0.6950	3m 09s
CatBoost	0.9755	0.7011	94.70	60.87	80.97	0.6949	1m 58s
H2O GBM	0.9741	0.6832	94.28	57.96	84.76	0.6884	59s
Lasso	0.9714	0.6432	94.06	56.99	82.78	0.6751	10m 06s
Elastic Net	0.9714	0.6432	94.05	56.96	82.77	0.6748	13m 25s
Logistic Regression	0.9714	0.6430	94.06	56.97	82.76	0.6749	1m 50s
Ridge	0.9711	0.6397	93.73	55.03	86.98	0.6741	7m 41s
Decision Tree	0.9682	0.5996	93.52	54.43	80.06	0.6480	17s
Naive Bayes	0.8736	0.2767	86.80	30.72	61.45	0.4096	5m 06s

Models ranked by ROC-AUC; small differences are descriptive, not significance tests. Time includes fit, prediction and output processing. Logistic regression uses a small stabilizing penalty; Lasso/Ridge/Elastic Net use 3-fold training CV.

Random-forest feature importance: top 8

Rank	Predictor	Raw importance	Share%
1	Entrepreneurial intention	33,416.22	45.54
2	Established business owner	11,858.05	16.16
3	Self-efficacy	2,092.05	2.85
4	Knowing entrepreneurs	2,046.47	2.79
5	Opportunity perception	1,432.19	1.95
6	Safety and security	1,337.17	1.82
7	Education	1,254.01	1.71
8	Living conditions	1,249.04	1.70

Share is the percentage of total forest impurity importance across all 51 encoded predictors, not explained variance. Raw values are not comparable across samples. Missing indicators are derived preprocessing features.

Leading Legatum predictors within the same forest

Rank	Legatum predictor	Raw importance	Share%
1	Safety and security	1,337.17	1.82
2	Education	1,254.01	1.71
3	Living conditions	1,249.04	1.70

Legatum subset ranked by raw importance; shares use total forest importance, not the Legatum subtotal.

Source: Completed Objective 2 batch, 28 September 2026; 02_obj2_global_iv_OME. Scores and importance are taken directly from saved CSV outputs.